

SUPPLEMENTARY MATERIAL

Accurate Detection of Convergent Mutations in Large Protein Alignments with ConDor

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Our MSAs, phylogenetic trees, scripts and results analysis are accessible from the Github repository <https://github.com/evolbioinfo/condor-analysis>.

Table of Contents

• Table S1: Method comparison with threshold variation on sedge PEPC dataset with the genotypic annotation	Page 2
• Table S2: Method comparison with threshold variation on sedge PEPC dataset with the phenotypic annotation	Page 3
• Table S3: ConDor results on the 12 tested convergent positions on sedge PEPC dataset with the genotypic annotation	Page 4
• Table S4: ConDor results on the 11 tested convergent positions on sedge PEPC dataset with the phenotypic annotation	Page 5
• Table S5: Method comparison with threshold variation on HIV reverse transcriptase	Page 6
• Table S6: Results of FUBAR and MEME on on HIV reverse transcriptase dataset	Page 7
• Table S7: ConDor results on the 29 DRMs tested on HIV reverse transcriptase dataset	Page 8
• Table S8: Method comparison with threshold variation on fish rhodopsin dataset	Page 9
• Table S9: ConDor results on the 10 tested convergent mutations on fish rhodopsin dataset	Page 10

Genotypic 66 mutations tested 12 convergent mutations	TP	FP	FN	TN	Type1	Recall	Precision	F1 score
PC 0.8	1	6	11	48	0.11	0.08	0.14	0.11
PC 0.9	0	2	12	52	0.04	0.00	0.00	0.00
OC 0.8	8	7	4	47	0.13	0.67	0.53	0.59
OC 0.9	8	7	4	47	0.13	0.67	0.53	0.59
PCOC 0.8	7	4	5	50	0.07	0.58	0.64	0.61
PCOC 0.9	7	3	5	51	0.06	0.58	0.70	0.64
FADE JTT 100	11	4	1	50	0.07	0.92	0.73	0.81
FADE JTT 1000	5	1	7	53	0.02	0.42	0.83	0.56
Emergence JTT 0.10	7	16	5	38	0.30	0.58	0.30	0.40
Emergence JTT 0.05	5	12	7	42	0.22	0.42	0.29	0.34
Correlation 2	8	6	4	48	0.11	0.67	0.57	0.62
Correlation 5	4	3	8	51	0.06	0.33	0.57	0.42
Correlation 10	3	0	9	54	0.00	0.25	1.00	0.40
ConDor Em: 0.1, Corr: 2	5	3	7	51	0.06	0.42	0.62	0.50

Table S1: Method comparison with threshold variation on sedge PEPC dataset with the genotypic annotation. We display for PCOC, FADE, ConDor and their sub-components, several performance indicators on the detection of convergent mutations with the genotypic annotation according to (Besnard et al. 2009). We display the results using the best substitution matrix (JTT). See text and Table 2 for details. PC 0.8 (0.9): mutations detected with a profile change in convergent clades, with posterior probability >0.8 (as used in (Rey et al. 2018)) (or >0.9). OC 0.8 (0.9): mutations detected on positions where changes occur on the branches leading to the convergent clades, with posterior probability >0.8 (or >0.9). PCOC 0.8 (0.9): combination of PC and OC components with posterior probability >0.8 (or >0.9). FADE 100 (1000): mutations with mutations showing evolution under directional selection in convergent clades, with Bayes Factor >100 (default) (or >1000). Emergence 0.10 (0.05): mutations showing a number of EEMs statistically higher than expected with p-value <0.1 (default) (or <0.05) (after Holm-Bonferroni correction for multiple tests). Correlation 2 (5, 10): mutations positively correlated with C4 annotation, with log Bayes factor >2 (or >5, >10) (corresponding to thresholds in the BayesTraits documentation). ConDor: combination of Emergence <0.10 and Correlation >2.

For ConDor we report in Table 2 the thresholds where Correlation and Emergence had the best F1 score (logBF >2 and p-value <0.10). For PCOC, we kept the thresholds used for the sedge dataset in the original publication (Rey et al. 2018), even though PCOC with a posterior probability of 0.9 has a slightly better F1 score. For FADE we kept a threshold of 100, which is the default value that seems to be best suited for small datasets like this one.

Phenotypic annotation 59 mutations tested 11 convergent mutations	TP	FP	FN	TN	Type1	Recall	Precision	F1 score
PC 0.8	0	4	11	44	0.08	0.00	0.00	0.00
PC 0.9	0	3	11	45	0.06	0.00	0.00	0.00
OC 0.8	0	1	11	47	0.02	0.00	0.00	0.00
OC 0.9	0	1	11	47	0.02	0.00	0.00	0.00
PCOC 0.8	0	0	11	48	0.00	0.00	0.00	0.00
PCOC 0.9	0	0	11	48	0.00	0.00	0.00	0.00
FADE JTT 100	8	9	3	39	0.19	0.73	0.47	0.57
FADE JTT 1000	5	4	6	44	0.08	0.45	0.56	0.50
Emergence JTT 0.1	6	18	5	30	0.38	0.55	0.25	0.34
Emergence JTT 0.05	4	13	7	35	0.27	0.36	0.24	0.29
Correlation 2	6	7	5	41	0.15	0.55	0.46	0.50
Correlation 5	3	2	8	46	0.04	0.27	0.60	0.38
Correlation 10	0	0	11	48	0.00	0.00	N/A	0.00
ConDor Em: 0.1, Corr: 2	4	3	7	45	0.06	0.36	0.57	0.44

Table S2: Method comparison with threshold variation on sedge PEPC dataset with the phenotypic annotation. We display for PCOC, FADE, ConDor and their sub-components, several performance indicators on the detection of convergent positions with the phenotypic annotation. We display the results using the best substitution matrix (JTT). See text and Table 2 for details. PC 0.8 (0.9): mutations detected with a profile change in convergent clades, with posterior probability >0.8 (as used in (Rey et al. 2018)) (or >0.9). OC 0.8 (0.9): mutations detected on positions where changes occur on the branches leading to the convergent clades, with posterior probability >0.8 (or >0.9). PCOC 0.8 (0.9): combination of PC and OC components with posterior probability >0.8 (or >0.9). FADE 100 (1000): mutations with mutations showing evolution under directional selection in convergent clades, with Bayes Factor >100 (default) (or >1000). Emergence 0.10 (0.05): mutations showing a number of EEMs statistically higher than expected with p-value <0.1 (default) (or <0.05) (after Holm-Bonferroni correction for multiple tests). Correlation 2 (5, 10): mutations positively correlated with C4 annotation, with log Bayes factor >2 (or >5, >10) (corresponding to thresholds in the BayesTraits documentation). ConDor: combination of Emergence <0.10 and Correlation >2.

For ConDor we report in Table 2 the thresholds where Correlation and Emergence had the best F1 score (logBF >2 and p-value <0.10). For PCOC, we kept the thresholds used for the sedge dataset in the original publication (Rey et al. 2018). For FADE we kept a threshold of 100, which is the default value that seems to be best suited for small datasets like this one.

Mutation	nbEEM	nbseq	evol_rate	substitution_rate	p-value	detected_EEM	log_BF_correlation	correlation	detected_correlation	detected_ConDor
H665N	7	25	4.26215	0.08952861	0.00330	PASS	12.562800	positive	PASS	PASS
A780S	5	22	1.84445	0.28945548	0.02030	PASS	18.139494	positive	PASS	PASS
E572Q	6	21	5.78303	0.19893438	0.02475	PASS	4.081248	positive	PASS	PASS
P540T	5	23	5.30366	0.05940552	0.03380	PASS	17.557648	positive	PASS	PASS
G839K	3	8	3.74791	0.02009760	0.04250	PASS	0.060904	positive	NOT PASS	NOT PASS
A573N	3	15	2.99025	0.04135078	0.05144	PASS	-0.049806	positive	NOT PASS	NOT PASS
M749T	3	4	4.96542	0.05271843	0.09224	PASS	2.012038	positive	PASS	PASS
F505L	4	17	3.84832	0.10013977	0.52510	NOT PASS	7.992726	positive	PASS	NOT PASS
Y751F	4	11	3.92156	0.17249877	0.52510	NOT PASS	3.733476	positive	PASS	NOT PASS
S761A	3	14	1.52217	0.25692931	0.52510	NOT PASS	1.075694	positive	NOT PASS	NOT PASS
I731V	5	20	2.94386	0.50330523	0.66683	NOT PASS	3.268858	positive	PASS	NOT PASS
E623D	4	14	5.42408	0.47239217	0.73763	NOT PASS	-3.900954	positive	NOT PASS	NOT PASS

Table S3: ConDor results on the 12 tested convergent mutations derived from (Besnard et al. 2009) on sedge PEPC dataset with the genotypic annotation. We display several statistics for the 12 convergent mutations considered as true positives in the sedge PEPC analysis. *nbEEM*: number of independent emergence-events of a mutation; *nbseq*: number of sequences the mutation is present; *evol_rate*: evolutionary rate of the site; *substitution_rate*: substitution rate of the mutation according to JTT model of evolution; *p-value*: adjusted *p*-value of the mutation after Holm-Bonferroni correction for the Emergence component; *detected_EEM*: PASS if the mutation is detected by the Emergence component (*p*-value <0.1); *log_BF_correlation*: log Bayes factor of the mutation for the Correlation component; *correlation*: positive if the mutation is positively correlated with the genotype (presence of A780S mutation, a major determinant of C4-specific characteristics (Bläsing et al. 2000)); *detected_correlation*: PASS if the mutation is detected by the Correlation component (*logBF* >2 and positive correlation); *detected_ConDor*: PASS if detected by both components.

For more information on these mutations (and others), see the MutationsValidationsReferences document available on GitHub (see URL above).

Mutation	nbEEM	nbseq	evol_rate	substitution_rate	p-value	detected_EEM	log_BF_correlation	correlation	detected_Correlation	detected_ConDor
E572Q	6	21	5.78628	0.19893438	0.00295	PASS	3.688428	positive	PASS	PASS
H665N	6	20	3.71715	0.08952861	0.00295	PASS	3.450806	positive	PASS	PASS
A573N	3	15	1.72855	0.04135078	0.00295	PASS	0.218842	positive	NOT PASS	NOT PASS
A780S	4	17	1.66748	0.28945548	0.04944	PASS	8.961452	positive	PASS	PASS
P540T	4	18	4.79661	0.05940552	0.06764	PASS	8.804466	positive	PASS	PASS
M749T	3	4	5.44982	0.05271843	0.07409	PASS	0.613974	positive	NOT PASS	NOT PASS
I731V	5	20	2.13330	0.50330523	0.40686	NOT PASS	2.093032	positive	PASS	NOT PASS
S761A	3	14	1.51020	0.25692931	0.40686	NOT PASS	1.157994	positive	NOT PASS	NOT PASS
F505L	4	17	4.46136	0.10013977	0.49110	NOT PASS	6.846750	positive	PASS	NOT PASS
Y751F	3	6	2.97287	0.17249877	0.56169	NOT PASS	0.496870	positive	NOT PASS	NOT PASS
E623D	4	14	5.16626	0.47239217	0.78432	NOT PASS	-4.961670	positive	NOT PASS	NOT PASS

Table S4: ConDor results on the 11 tested convergent mutations derived from (Besnard et al. 2009) on sedge PEPC dataset with the phenotypic annotation. We display several statistics for the 11 convergent mutations considered as true positives in the sedge PEPC analysis. *nbEEM*: number of independent emergence-events of a mutation; *nbseq*: number of sequences the mutation is present; *evol_rate*: evolutionary rate of the site; *substitution_rate*: substitution rate of the mutation according to JTT model of evolution; *p-value*: adjusted *p*-value of the mutation after Holm-Bonferroni correction for the Emergence component; *detected_EEM*: PASS if the mutation is detected by the Emergence component (*p*-value <0.1); *log_BF_correlation*: log Bayes factor of the mutation for the Correlation component; *correlation*: positive if the mutation is positively correlated with the phenotype (C4 metabolism); *detected_correlation*: PASS if the mutation is detected by the Correlation component (*logBF* ≥2 and positive correlation); *detected_ConDor*: PASS if detected by both components.

For more information on these mutations (and others), see the MutationsValidationsReferences document available on GitHub (see URL above).

HIV dataset – 29 DRMs 240 mutations tested 29 mutations searched	TP	FP	FN	TN	Type1	Recall	Precision	F1 score
PC 0.8	10	10	19	201	0.05	0.35	0.50	0.41
PC 0.9	9	10	20	201	0.05	0.31	0.47	0.38
FADE HIVb 100	22	66	7	145	0.31	0.76	0.25	0.38
FADE HIVb 1000	21	51	8	160	0.24	0.72	0.29	0.42
FADE HIVb INFINITY	13	6	16	205	0.03	0.45	0.68	0.54
FADE JTT 100	24	69	5	142	0.33	0.83	0.26	0.39
FADE JTT 1000	24	54	5	157	0.26	0.83	0.31	0.45
FADE JTT INFINITY	14	8	15	203	0.04	0.48	0.64	0.55
Emergence HIVb 0.10	20	67	9	144	0.32	0.69	0.23	0.34
Emergence HIVb 0.05	18	64	11	147	0.30	0.62	0.22	0.32
Emergence JTT 0.10	21	78	8	133	0.37	0.72	0.21	0.33
Emergence JTT 0.05	21	75	8	136	0.36	0.72	0.22	0.34
Correlation 2	23	26	6	185	0.12	0.79	0.47	0.59
Correlation 5	22	22	7	189	0.10	0.76	0.50	0.60
Correlation 10	19	12	10	199	0.06	0.66	0.61	0.63
Correlation 20	16	3	13	208	0.01	0.55	0.84	0.67
ConDor Em: 0.1, Corr: 20	15	2	14	209	0.01	0.52	0.88	0.65

Table S5: Method comparison with threshold variation on HIV reverse transcriptase dataset. We display for PC, FADE, ConDor, and ConDor sub-components, several performance indicators of the detection of convergent mutations. We display the results using the best substitution matrix (HIVb) and, when possible, with JTT, which is a form of model misspecification. See text and Table 3 for details. PC 0.8 (0.9): mutations detected on positions with a profile change in convergent clades with a posterior probability >0.8 (as used in (Rey et al. 2018)) (or >0.9). We do not display results for OC or PCOC as no mutations were detected by those methods. FADE 100 (1000, INFINITY): mutations showing evolution under directional selection, with Bayes Factor >100 (default) (>1000 or INFINITY, i.e., >10¹⁶). Emergence 0.1 (0.05): mutations showing a number of EEMs statistically higher than expected at a p-value <0.1 (default) (or <0.05) after Holm-Bonferroni correction for multiple testing. Correlation 2 (5, 10, 20): mutations positively correlated with the treatment status, with log Bayes factor >2 (or >5, >10 and >20) (corresponding to thresholds in the BayesTraits documentation). ConDor: combination of Emergence <0.1 and Correlation >20.

For PC we report in Table 3 the default threshold which also exhibits the best F1 score. For FADE we report the thresholds for INFINITY (best F1 score). For ConDor we report in Table 3 the threshold where Correlation had the best F1 score (logBF >20) and the default value for Emergence which is nearly optimal (p-value <0.1). JTT gives better results for FADE and Emergence depending on the threshold but the difference is not significant which demonstrates the robustness of the methods to model change.

95 tested positions	TP	FP	FN	TN	Type1	Recall	Precision	F1 score
FUBAR 0.9	0	9	24	62	0.13	0.00	0.00	0.00
FUBAR 0.8	0	10	24	61	0.14	0.00	0.00	0.00
MEME 0.1	4	32	20	39	0.45	0.17	0.11	0.13
MEME 0.05	3	30	21	41	0.42	0.12	0.09	0.11

Table S6: Results of FUBAR and MEME on the detection of DRMs on HIV reverse transcriptase dataset. We applied MEME (Mixed Effect of Molecular Evolution; (Murrell, Wertheim, et al. 2012)) and FUBAR (Fast, Unconstrained Bayesian AppRoximation; (Murrell et al. 2013)) from the Hyphy software suite (Pond, Frost, and Muse 2005) to the HIV dataset to assess whether codons corresponding to DRMs appeared under positive selection. FUBAR is a Bayesian approach that calculates site-by-site non-synonymous (dN) and synonymous (dS) substitution rates. MEME and FUBAR are widely used to detect positive selection and were previously applied for the detection of mutations on the fish rhodopsin as well as on the reverse transcriptase of HIV (Murrell, Wertheim, et al. 2012; Murrell et al. 2013). We present results by position and not by mutation. We used default parameters from the hyphy command line package and varied the detection thresholds for both methods using <http://vision.hyphy.org> website. The selection pressures for each site are assumed to be constant throughout the phylogeny. MEME aims to detect whether positive selection at some sites applies only to some branches. Selection pressures for each site are thus no longer assumed to be constant throughout the phylogeny but may be episodic. A priori, the DRMs should not correspond to constant pressures throughout the phylogeny. Indeed, DRMs appear in response to treatment, which should result in episodic selection pressures. On the HIV dataset, with default options at a posterior probability of 0.9, FUBAR detects 9 sites under positive selection, none of which show resistance mutations, which is an expected result. MEME, with default parameters, detects 4 positions with DRMs among 36 positions with a p-value <0.1. Even though DRMs are mutations resulting from episodic pressures, it appears on this dataset that phenotype information is essential for the detection of DRMs. Indeed, these methods do not allow annotation of branches under positive selection (sometimes called foreground branches) and thus DRMs are difficult to detect. We do not report results for PAML (Yang, 2000) which is not scalable for such big dataset. PAML was indeed previously reported to run for more than 35 hours on 349 taxa and to report an error regarding too many ambiguities in a dataset with 476 taxa (Murrell et al. 2013).

Mutation	nbEEM	nbseq	evol_rate	substitution_rate	p-value	detected_EEM	log_BF_correlation	correlation	detected_correlation	detected_ConDor
M184V	225	383	3.26559	0.07187959	0.0011999	PASS	710.580732	positive	PASS	PASS
K103N	132	183	3.26559	0.23891621	0.0011999	PASS	210.261494	positive	PASS	PASS
Y181C	102	139	3.26559	0.11400647	0.0011999	PASS	174.214600	positive	PASS	PASS
G190A	74	95	3.26559	0.08192216	0.0011999	PASS	93.0463600	positive	PASS	PASS
M41L	39	54	1.05893	0.05579457	0.0011999	PASS	65.047254	positive	PASS	PASS
D67N	45	50	1.05893	0.39458277	0.0011999	PASS	64.324638	positive	PASS	PASS
T215Y	35	54	1.05893	0.00300665	0.0011999	PASS	61.723256	positive	PASS	PASS
T215F	36	49	1.05893	0.00040202	0.0011999	PASS	60.838252	positive	PASS	PASS
H221Y	43	53	1.05893	0.22112537	0.0011999	PASS	51.111274	positive	PASS	PASS
A98G	34	40	1.05893	0.06853285	0.0011999	PASS	42.397120	positive	PASS	PASS
K101E	40	47	1.05893	0.13948574	0.0011999	PASS	41.757866	positive	PASS	PASS
L210W	24	28	1.05893	0.03927488	0.0011999	PASS	34.011340	positive	PASS	PASS
K219Q	26	29	1.05893	0.19789825	0.0011999	PASS	30.773352	positive	PASS	PASS
Y188L	20	21	1.05893	0.00168447	0.0011999	PASS	24.616342	positive	PASS	PASS
P225H	18	20	1.05815	0.05990573	0.0011999	PASS	19.511092	positive	NOT PASS	NOT PASS
V90I	23	33	0.16748	0.57999971	0.0011999	PASS	8.388428	positive	NOT PASS	NOT PASS
E138A	46	60	1.78041	0.05624911	0.0011999	PASS	-14.818514	positive	NOT PASS	NOT PASS
K70R	56	64	1.05893	0.61811372	0.0174283	PASS	83.476558	positive	PASS	PASS
M230L	9	14	0.16733	0.05579458	0.0639836	PASS	-1.199256	positive	NOT PASS	NOT PASS
K219E	20	22	1.05893	0.13948575	0.0937656	PASS	13.802398	positive	NOT PASS	NOT PASS
F227L	11	12	0.16733	0.13030577	0.1507349	NOT PASS	5.596744	positive	NOT PASS	NOT PASS
V108I	47	51	1.05893	0.57999971	0.3198330	NOT PASS	67.390418	positive	PASS	NOT PASS
V106M	10	11	1.05893	0.22407556	0.4296070	NOT PASS	1.528008	positive	NOT PASS	NOT PASS
V179D	13	14	3.26559	0.03426363	0.6130637	NOT PASS	-4.923794	positive	NOT PASS	NOT PASS
V75I	20	23	1.05893	0.57999971	1.0	NOT PASS	10.251330	positive	NOT PASS	NOT PASS
V106A	12	12	1.05893	0.24896021	1.0	NOT PASS	9.773664	positive	NOT PASS	NOT PASS
K65R	10	11	0.16733	0.61811372	1.0	NOT PASS	4.122492	positive	NOT PASS	NOT PASS
V106I	30	38	1.05893	0.57999971	1.0	NOT PASS	-6.853070	positive	NOT PASS	NOT PASS
V179T	11	15	3.26559	0.02318920	1.0	NOT PASS	-6.885248	positive	NOT PASS	NOT PASS

Table S7: ConDor results on the 29 DRMs found in ≥ 10 sequences on HIV reverse transcriptase dataset. We display several statistics for the 29 DRMs considered as true positives in the HIV analysis. *nbEEM*: number of independent emergence-events of a mutation; *nbseq*: number of sequences the mutation is present; *evol_rate*: evolutionary rate of the site; *substitution_rate*: substitution rate of the mutation according to HIVb model of evolution; *p-value*: adjusted *p*-value of the mutation after Holm-Bonferroni correction for the Emergence component; *detected_EEM*: PASS if the mutation is detected by the Emergence component (*p*-value < 0.1); *log_BF_correlation*: log Bayes factor of the mutation for the Correlation component; *correlation*: positive if the mutation is positively correlated with the phenotype (treated patient); *detected_correlation*: PASS if the mutation is detected by the Correlation component ($\log BF > 20$ and positive correlation); *detected_ConDor*: PASS if detected by both components.

For more information on these mutations (and others), see the *MutationsValidationsReferences* document available on GitHub (see URL above).

Marine and Brackish/Fresh 358 mutations tested 10 mutations searched	TP	FP	FN	TN	Type1	Recall	Precision	F1 score
PC 0.8	3	24	7	324	0.07	0.3	0.11	0.16
PC 0.9	3	22	7	326	0.06	0.3	0.12	0.17
FADE LG 100	6	103	4	245	0.30	0.6	0.06	0.10
FADE LG 1000	6	72	4	276	0.21	0.6	0.08	0.14
FADE JTT 100	6	101	4	247	0.29	0.6	0.06	0.10
FADE JTT 1000	4	81	6	254	0.24	0.4	0.05	0.08
Emergence LG 0.1	4	56	6	292	0.16	0.4	0.07	0.11
Emergence LG 0.05	4	50	6	298	0.14	0.4	0.07	0.12
Emergence JTT 0.1	4	72	6	276	0.21	0.4	0.05	0.09
Emergence JTT 0.05	4	62	6	286	0.18	0.4	0.06	0.11
Correlation 2	7	136	3	212	0.39	0.7	0.05	0.09
Correlation 5	7	114	3	234	0.33	0.7	0.06	0.11
Correlation 10	6	95	4	253	0.27	0.6	0.06	0.11
Correlation 20	6	67	4	281	0.19	0.6	0.09	0.14
ConDor (Em. 0.1, Corr. 20)	4	14	6	334	0.04	0.4	0.22	0.29
ConDor (Em. 0.05, Corr. 20)	4	13	6	335	0.04	0.4	0.24	0.30

Table S8: Method comparison with threshold variation on fish rhodopsin dataset. We display for PC, FADE, ConDor, and ConDor sub-components, several performance indicators of the detection of convergent mutations. We display the results using the second-best model (LG) for comparison purpose and, when possible, with JTT, which is a form of model misspecification. See text and Table 4 for details. PC (profile change) 0.8 (0.9): mutations detected on positions with a profile change in convergent clades with a posterior probability >0.8 (or >0.9). We do not display results for OC or PCOC as no mutations were detected by those methods. FADE 100 (1000): mutations showing evolution under directional selection, with Bayes Factor >100 (default) (or >1000). Emergence 0.1 (0.05): mutations showing a number of EEMs statistically higher than expected at a p-value <0.1 (default) (or <0.05) after Holm-Bonferroni correction for multiple testing. Correlation 2 (5, 10, 20): mutations positively correlated with the treatment status, with log Bayes factor >2 (or >5, >10 and >20) (corresponding to thresholds in the BayesTraits documentation). ConDor: combination of Emergence < 0.1 (or <0.05) and Correlation >20.

In Table 4: we report the results of PC with 0.8 default threshold, which is nearly optimal; a Bayes Factor >1000 for FADE; a default p-value <0.1 for Emergence, which is nearly optimal; a log BF >20 for Correlation; and for ConDor, the default combination of Emergence with p-value <0.1 and Correlation with logBF >20, which is nearly optimal.

Mutation	nbEEM	nbseq	evol_rate	substitution_rate	p-value	detected_EEM	log_BF_correlation	correlation	detected_correlation	detected_ConDor
Marine										
D83N	31	541	2.63353	0.25549607	0.017898	PASS	102.673688	positive	PASS	PASS
A292S	44	135	1.54265	0.379624936	0.017898	PASS	58.78318	positive	PASS	PASS
Y261F	3	1683	0.81975	0.293218663	1.0	NOT PASS	43.600086	positive	PASS	NOT PASS
E122Q	3	41	0.65273	0.254405918	1.0	NOT PASS	5.603954	positive	NOT PASS	NOT PASS
Fresh/Brack										
N83D	33	1163	2.63353	0.209768051	0.017898	PASS	134.251012	positive	PASS	PASS
F261Y	20	364	0.81975	0.373725726	0.017898	PASS	43.31748	positive	PASS	PASS
S292A	11	1802	1.54265	0.298826113	1.0	NOT PASS	72.839972	positive	PASS	NOT PASS
E122I	3	48	0.65273	0.002727632	1.0	NOT PASS	1.581026	positive	NOT PASS	NOT PASS
QI122E	3	1597	0.65273	0.148888295	1.0	NOT PASS	0.425834	positive	NOT PASS	NOT PASS
Y102F	3	11	0.15045	0.293218663	1.0	NOT PASS	-12.38451	positive	NOT PASS	NOT PASS

Table S9: ConDor results on the 10 tested convergent mutations on fish rhodopsin dataset using marine and fresh/brackish phenotype. We display several statistics for the 10 mutations considered as true positives in the fish rhodopsin analysis. *nbEEM*: number of independent emergence-events of a mutation; *nbseq*: number of sequences the mutation is present; *evol_rate*: evolutionary rate of the site; *substitution_rate*: substitution rate of the mutation according to LG model of evolution; *p-value*: adjusted *p*-value of the mutation after Holm-Bonferroni correction for the Emergence component; *detected_EEM*: PASS if the mutation is detected by the Emergence component (*p*-value <0.1); *log_BF_correlation*: log Bayes factor of the mutation for the Correlation component; *correlation*: positive if the mutation is positively correlated with the phenotype; *detected_correlation*: PASS if the mutation is detected by the Correlation component (*logBF* ≥20 and positive correlation); *detected_ConDor*: PASS if detected by both components. The phenotype is indicated by the mention Marine or Fresh/brackish in the Mutation column.

For more information on these mutations (and others), see the MutationsValidationsReferences document available on GitHub (see URL above).